

Energy-Optimal Urban Greening in Desalination-Dependent Cities

A remote sensing and machine learning framework for balancing cooling benefits and irrigation energy

1. Data & Input

Study Area

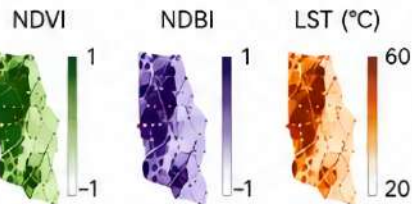
Jeddah, Saudi Arabia



Landsat 8
via Google Earth Engine



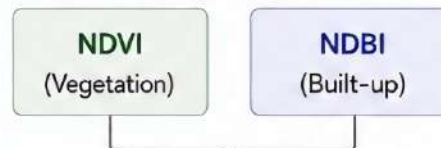
Remote Sensing Products



Hot-arid,
desalination-dependent
city

2. Modeling (XGBoost)

Input Variables



XGBoost
Model

Output

Predicted LST (°C)

Model Performance

$R^2 = 0.81$ | $RMSE = 3.71^\circ C$
5-fold CV $R^2 \approx 0.81$

FVC Scenarios

5–50%
(5% increments)



Increasing Fractional Vegetation Cover (FVC)

3. Energy Assessment

E_{cool} (Cooling Benefit)

Lower LST → Lower Cooling Demand

$$E_{cool} = \alpha \cdot \Delta LST$$



E_{desal} (Desalination Energy)

Irrigation Water from
Seawater Desalination

$$E_{desal} = w_0 \cdot \text{Irrigation Demand}$$



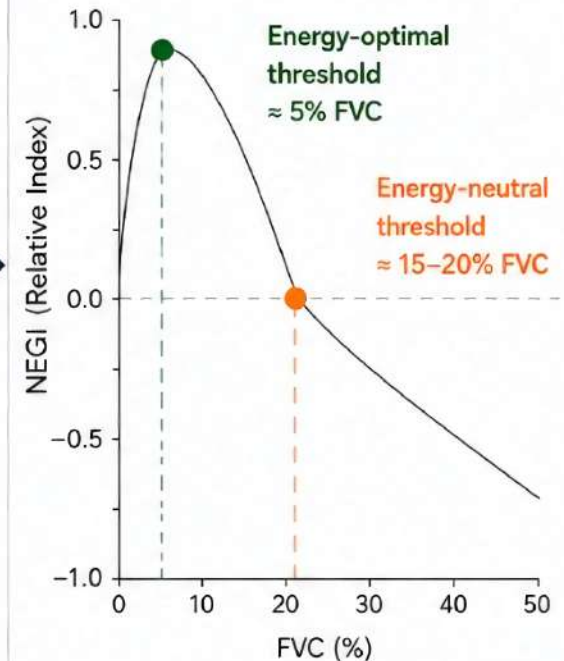
NEGI (Net Energy Gain Index)

$$NEGI = E_{cool} - E_{desal}$$



4. Key Findings

NEGI vs. Fractional Vegetation Cover (FVC)



- Maximum NEGI $\approx 5\%$ FVC.
- Energy-neutral threshold $\approx 15\text{--}20\%$ FVC.



Takeaway: Evidence-based identification of energy-optimal urban greening for desalination-dependent cities.

